DP Biology – B1.1

Carbohydrates and lipids

- How do differences in form give carbohydrates and lipids different functions?
- In what ways do lipids and carbohydrates differ as energy stores?

SL statements B1.1.1–13. Textbook pp. 181–204.

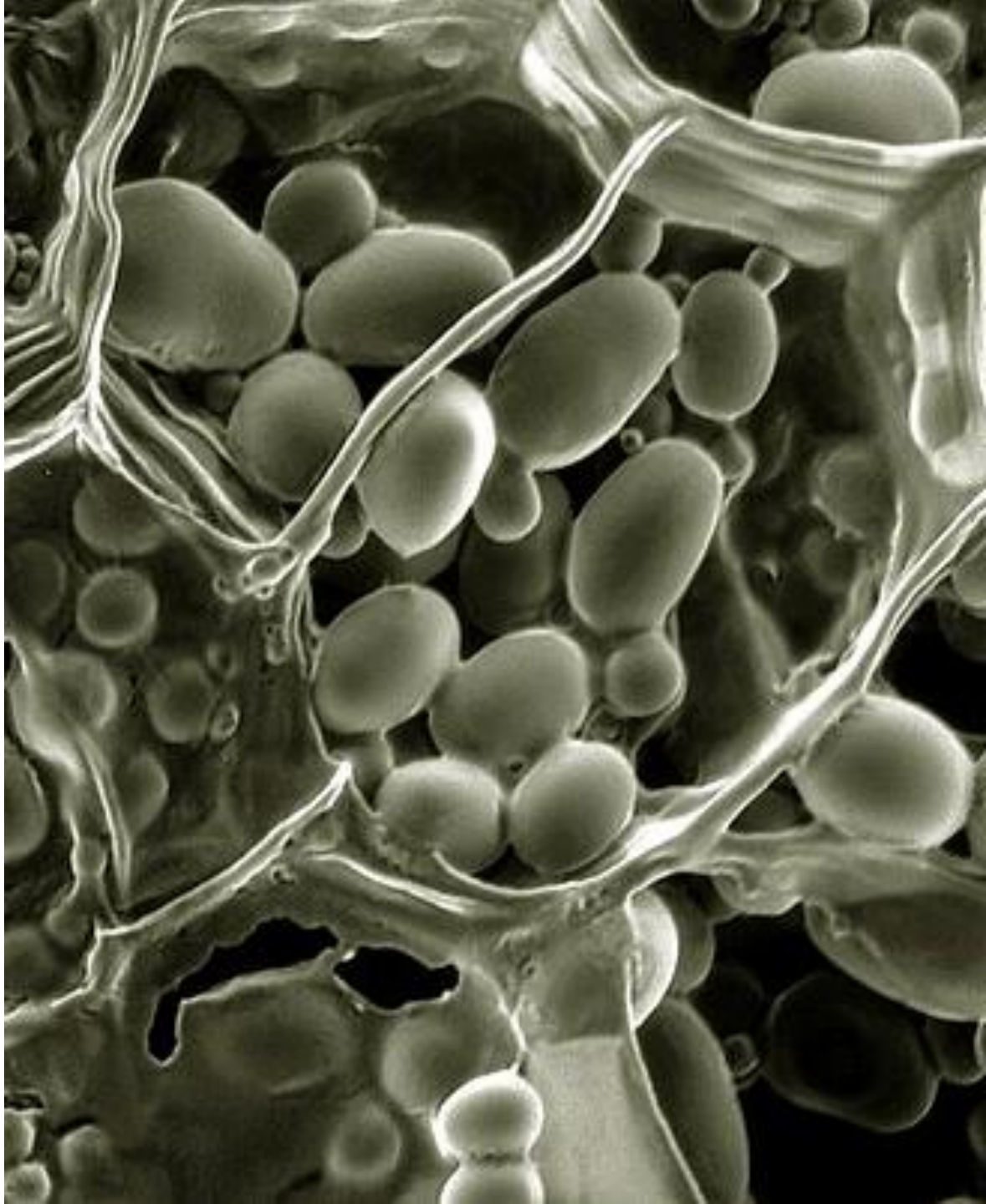


Image: starch grains from a potato, electron micrograph. Philippa Uwins, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	B1.1.1–4	Explain why carbon forms so many compounds, how condensation and hydrolysis work, and why glucose is so useful
2	B1.1.5–7	Relate starch, glycogen and cellulose to their functions, and explain glycoproteins in recognition
3	B1.1.8–13	Explain how triglycerides and phospholipids form and what they do, and why steroids cross membranes

Small units, big molecules, different jobs



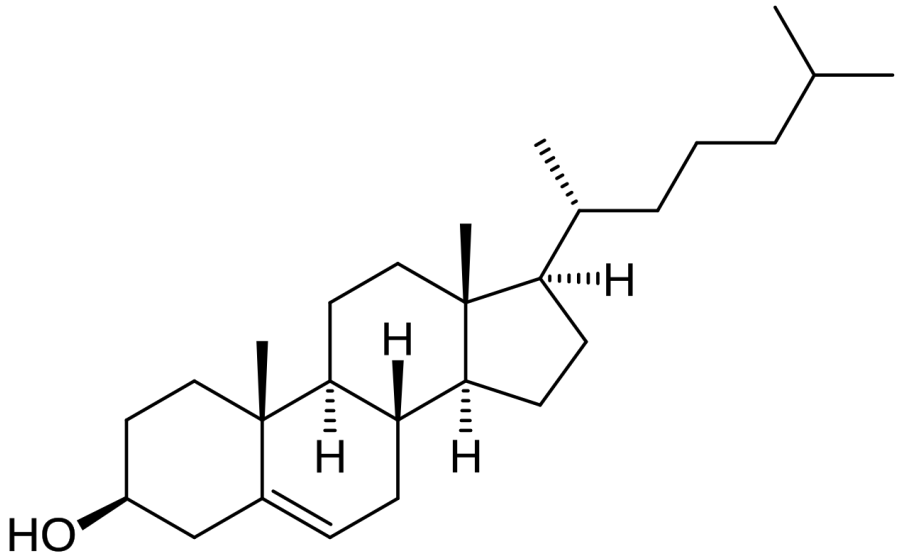
Living things build huge molecules from a few small ones: **carbohydrates** from sugars, many **lipids** from fatty acids and glycerol. Their shape and how they interact with water decide what each one does.

Big question

Starch, glycogen and cellulose are all glucose. Why is one a plant store, one an animal store, and one the strongest plant material?

Why carbon is the basis of life

B1.1.1



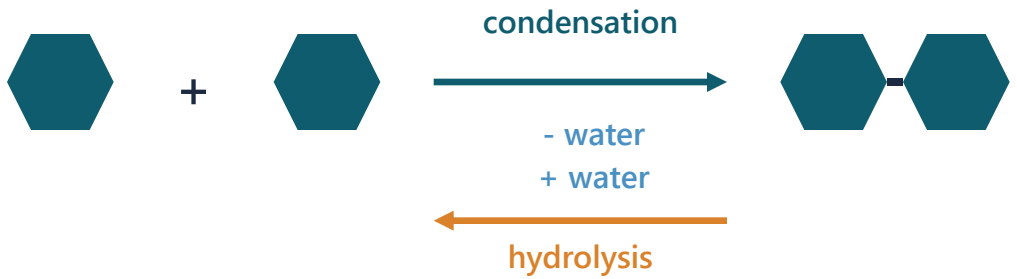
Cholesterol: rings and a branched chain. BorisTM, public domain.

Carbon forms **four covalent bonds**, to other carbons and to H, O, N, P and S. Covalent bonds are strong and stable.

So carbon builds **chains**, **branches** and **rings**: an almost endless variety of molecules.

Shape	Example
Chain	Fatty acids, cellulose
Branched	Amylopectin, glycogen
Ring	Glucose; steroids have four

Condensation and hydrolysis



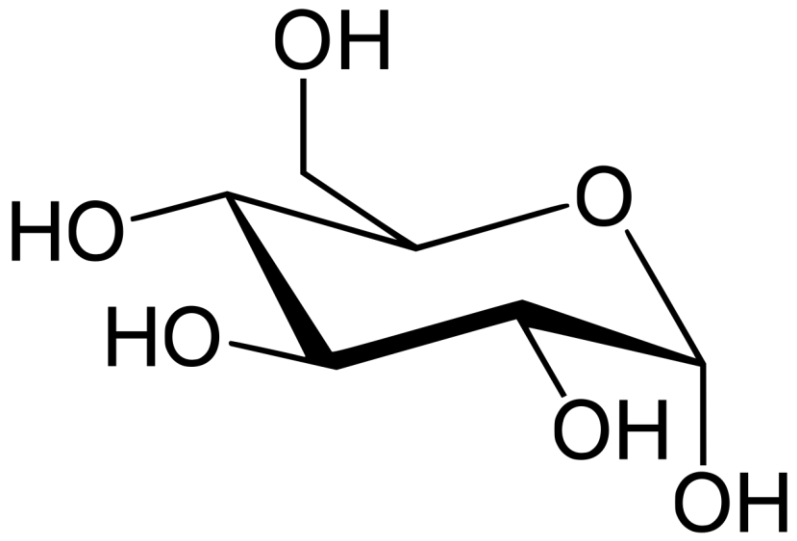
Condensation joins monomers by a covalent bond and **releases water**. Repeating it builds a **polymer**.

Hydrolysis uses water to break the bond: the reverse. **Digestion** is hydrolysis by enzymes.

Monomers	Polymer	Bond
Monosaccharides	Polysaccharide	Glycosidic
Amino acids	Polypeptide	Peptide
Fatty acids + glycerol	Triglyceride	Ester

Monosaccharides: single sugar units

B1.1.4



Alpha-glucose, a hexose. Yikrazuul, public domain.

Monosaccharides are single sugars: soluble and sweet.

Pentoses have 5 carbons (ribose); **hexoses** have 6 (glucose, fructose). Count the carbons: in a hexose ring, five carbons sit with an oxygen and the sixth sticks out.

Try it

Count the carbons in the glucose diagram. Find the one outside the ring.

Why glucose is so useful

B1.1.4

Soluble

Many -OH groups bond with water.

Stable

The ring does not react easily, so it can be stored and moved.

Energy

Its bonds release energy when respired.

Easily transported

Small and soluble, so blood plasma and phloem sap carry it to every cell.

Link

C1.2 Cell respiration: glucose is the main respiratory substrate.

Practice: food tests

B1.1.2–4

Test	Positive result	Shows
Iodine	Blue-black	Starch
Benedict's, heated	Green to brick red	Reducing sugar
Ethanol emulsion	Cloudy white	Lipid

1. A food turns iodine blue-black and Benedict's orange. What does it contain?
2. Starch is boiled with acid, then tested with Benedict's: brick red. Explain.

Practice: build and break

B1.1.1–4

Count water

Three glucose join in a chain. How many waters are released?

Name it

Name the bond between two monosaccharides.

Classify

A sugar has 5 carbons. Pentose or hexose?

Then

Explain why digestion must happen before food can be absorbed.

Check-in: class 1

B1.1.1–4

1. State how many covalent bonds carbon forms.
2. Distinguish between condensation and hydrolysis.
3. Give the bond formed between amino acids.
4. Identify a six-carbon sugar as a pentose or hexose.
5. Give two properties of glucose and why each matters.

How to use this

Answer without your notes, then check against slides 3–9.

Starch: the energy store of plants

B1.1.5



amylose: unbranched, coils into a helix



amylopectin: branched

Starch is a polymer of **alpha-glucose**. Amylose coils; amylopectin branches every 20–30 units.

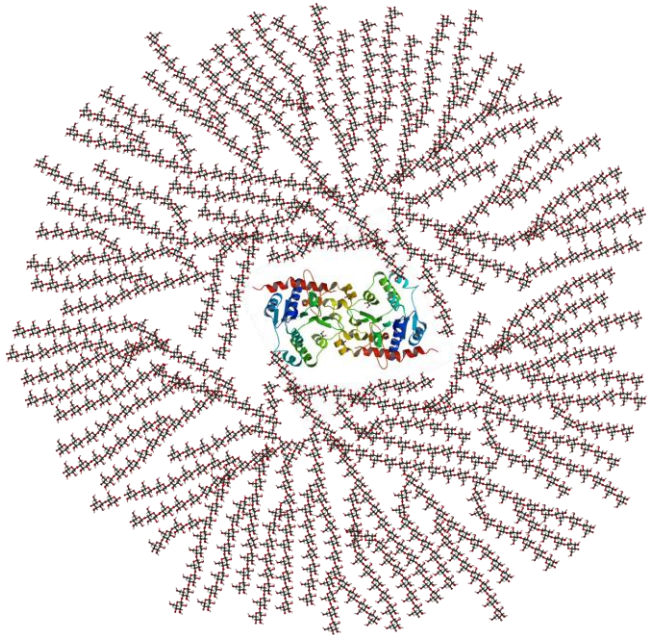
Coiled and branched, starch is **compact**; being huge, it is **insoluble** and has no osmotic effect.

Where

Grains in chloroplasts, seeds and storage organs such as potatoes.

Glycogen: the store of animals and fungi

B1.1.5



Branched glycogen. Mikael Haggstrom, public domain.

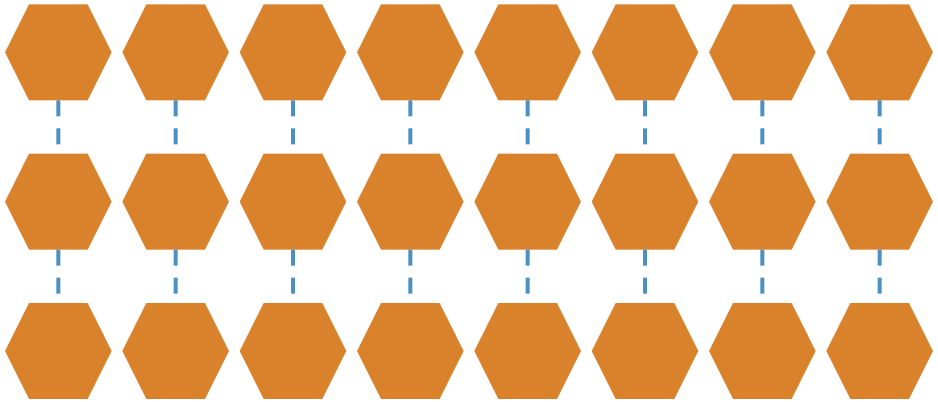
	Starch	Glycogen
Found in	Plants	Animals, fungi
Monomer	Alpha-glucose	Alpha-glucose
Branching	Some	Very highly branched
Stored in	Seeds, tubers	Liver, muscle

Key idea

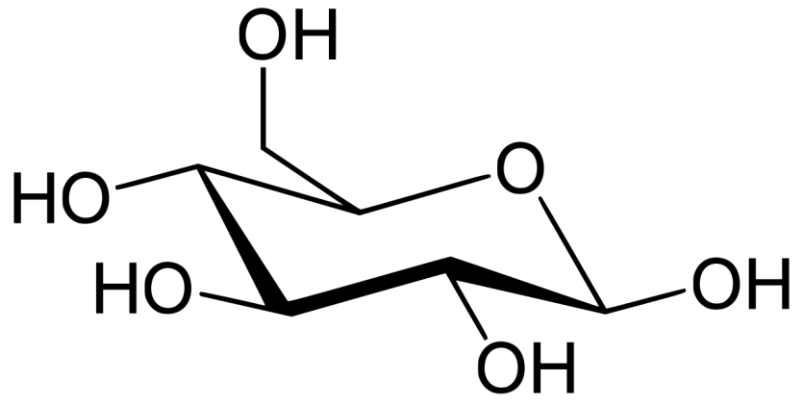
Every branch end is a place to add or remove glucose, so glycogen releases glucose fast: ideal for active animals.

Cellulose: a structural polysaccharide

B1.1.6



straight beta-glucose chains linked by hydrogen bonds (blue)



Beta-glucose. Yikrazuul, public domain.

Cellulose is made of **beta-glucose**: each unit is flipped relative to the next, so the chain is **straight**. Straight chains lie side by side, joined by **hydrogen bonds** into strong **microfibrils** in cell walls.

Key idea

Alpha or beta glucose changes the shape of the polymer, and so its job.

Same monomer, different polymer

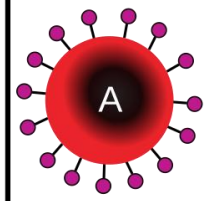
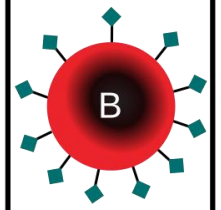
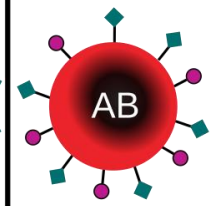
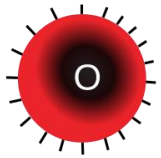
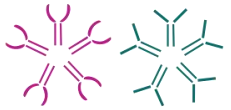
B1.1.5–6

	Starch (amylose)	Cellulose
Monomer	Alpha-glucose	Beta-glucose
Chain shape	Coiled helix	Straight
Between chains	No links	Hydrogen bonds: microfibrils
Function	Energy store	Strong cell walls
Digested by humans?	Yes, by amylase	No

Big question

Alpha or beta glucose decides the shape; the shape decides the job.

Glycoproteins: name tags on cells

	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cell	A antigen	 B antigen	 A and B antigens	None

ABO antigens on red blood cells. InvictaHOG, public domain.

A **glycoprotein** is a protein with a short sugar chain, in the membrane with the sugar facing **out**.

The sugar chains vary, so cells **recognize** each other. The **ABO antigens** differ only in the sugar at the end of the chain.

Use

The immune system uses these tags to tell self from non-self.

ABO blood groups and transfusions

B1.1.7

Group	Antigens on cells	Antibodies in plasma	Can receive from
A	A	Anti-B	A, O
B	B	Anti-A	B, O
AB	A and B	Neither	A, B, AB, O
O	Neither	Anti-A, anti-B	O only

Explain it

Group A blood given to a group B person: anti-A antibodies bind the A antigens and the cells clump.

Practice: structure and function

B1.1.5–7

Fast release

Why do muscles store glycogen, not cellulose?

No osmosis

Why store starch rather than glucose?

Transfusion

Can a group O person receive group A blood?

Explain

Use structure in each answer.

Check-in: class 2

B1.1.5–7

1. State the monomer of starch and of cellulose.
2. Explain two features of glycogen that suit energy storage.
3. Explain why cellulose forms strong fibres.
4. Outline how glycoproteins allow recognition.
5. Explain what happens if mismatched blood is transfused.

How to use this

Answer without your notes, then check against slides 11–17.

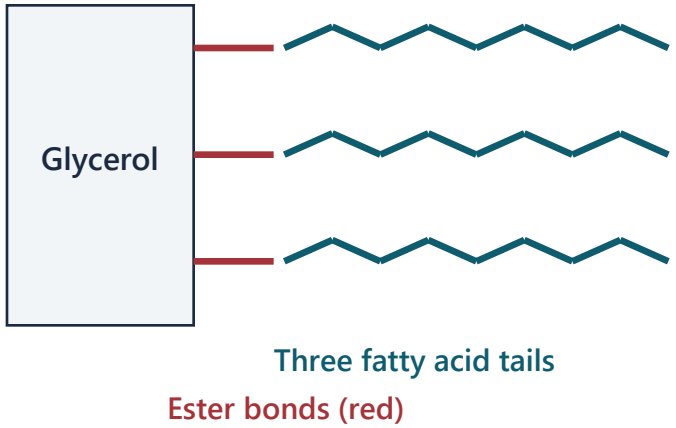
Lipids: defined by not dissolving in water

B1.1.8

Lipids dissolve in **non-polar solvents** (such as ethanol) but hardly in water, because they are mostly C-H, which is non-polar.

Lipid	Structure	Use
Fats and oils	Triglycerides	Energy store; insulation
Phospholipids	Glycerol, 2 fatty acids, phosphate	Membranes
Waxes	Long fatty acids and alcohols	Waterproofing
Steroids	Four fused rings	Hormones; cholesterol

Forming a triglyceride



Three fatty acids join one **glycerol** by condensation. Each join is an **ester bond** and releases water: **three** waters in all.

glycerol + 3 fatty acids → triglyceride + 3 water

Activity

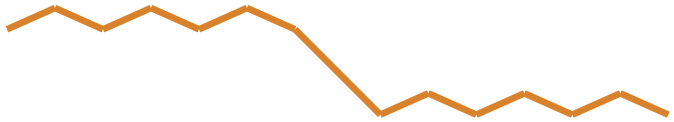
Draw a simplified triglyceride: a rectangle for glycerol and a zigzag for each fatty acid, with the ester bonds marked.

Saturated and unsaturated fatty acids

B1.1.10



saturated: straight



unsaturated (cis): kinked at C=C

Saturated: only single C-C bonds; straight; pack closely: **higher melting point**, solid fats.

Unsaturated: one (mono) or more (poly) **C=C**; in cis form each one kinks the chain: **lower melting point**, liquid oils.

Why it matters

Plants store oils that stay liquid in the cold. Mammals and birds store fats, liquid at body temperature but solid at room temperature.

Working with data: double bonds and melting point

B1.1.10

All four fatty acids have 18 carbons. Melting points are published values, rounded.

Fatty acid	C=C bonds	Melting point / °C
Stearic	0	70
Oleic	1	13
Linoleic	2	-5
Linolenic	3	-11

1. Describe the relationship.
2. Explain it.
3. Which are liquid in a fish at 4 °C?
4. Suggest why cold-climate seeds hold unsaturated oils.

Triglycerides in adipose tissue



Walruses with thick blubber. Andrew Trites, NOAA, CC BY 2.0.



Adipose tissue. Chalova.lina, CC BY 4.0.

Adipose tissue stores triglycerides: a **long-term energy store**, about twice the energy per gram of carbohydrate.

Fat conducts heat poorly: a **thermal insulator**.

Link

Blubber is thickest in endotherms in cold water, which conducts heat away fast (A1.1).

Lipids or carbohydrates for storage?

B1.1.11

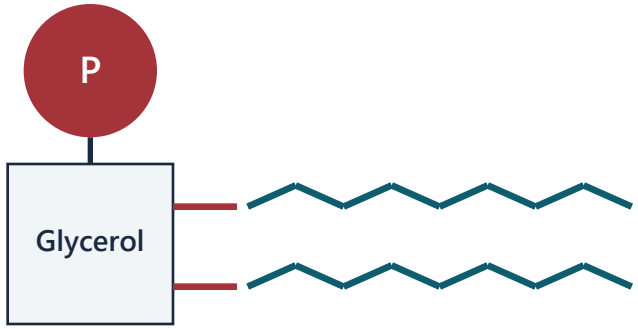
	Triglycerides	Glycogen, starch
Energy per gram	About 37 kJ	About 17 kJ
Mass for same energy	Low: good for moving animals	About twice as heavy
Release	Slow	Fast
Respiration	Aerobic only	Aerobic or anaerobic
Use	Long term	Short term

Guiding question

Lipids store more per gram; carbohydrates release energy faster and without oxygen.

Forming a phospholipid

B1.1.9



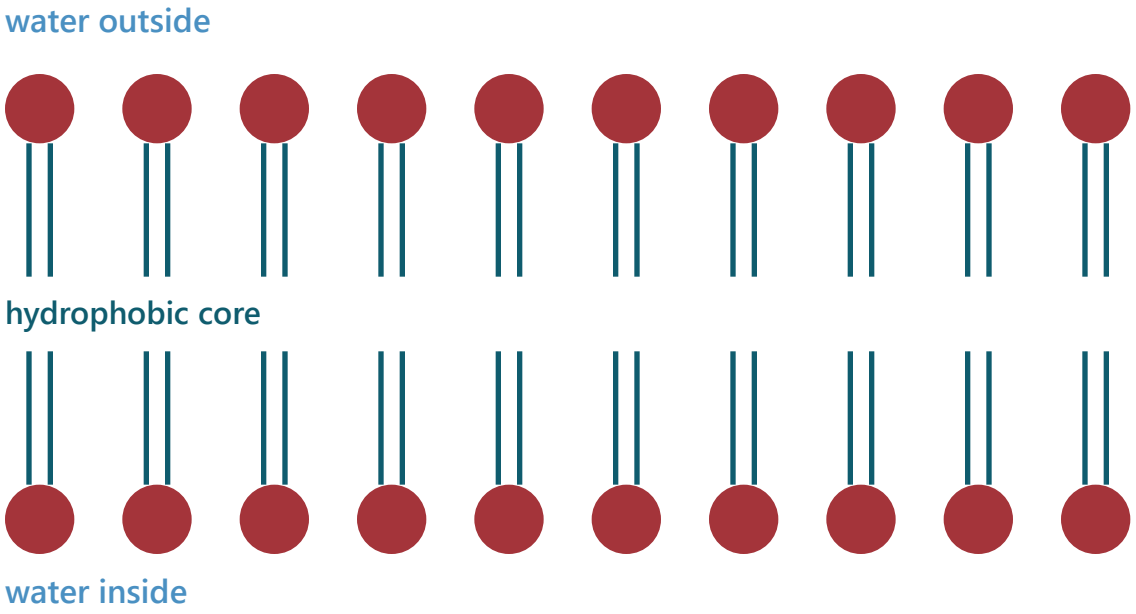
phosphate head, glycerol, two tails

	Triglyceride	Phospholipid
Fatty acids	Three	Two
Phosphate	None	One
Role	Energy store	Membranes

The phosphate head is charged and **hydrophilic**; the tails are **hydrophobic**. A molecule with both is **amphipathic**.

Phospholipids form bilayers

B1.1.12

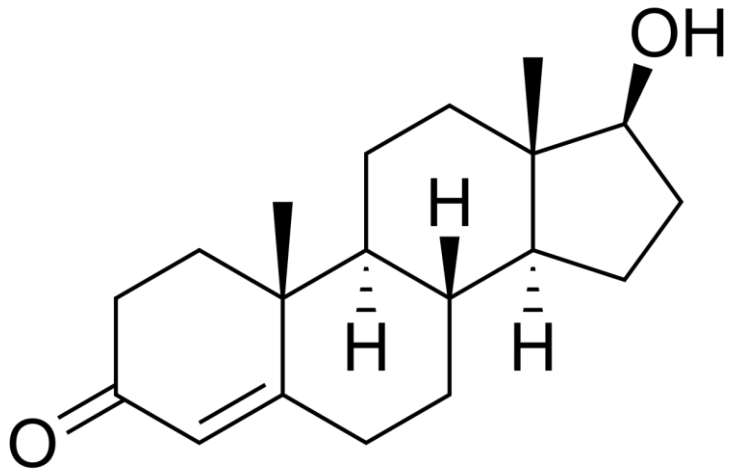


In water, heads face the water and tails hide in the middle: a **bilayer**. It forms by itself, with no energy input.

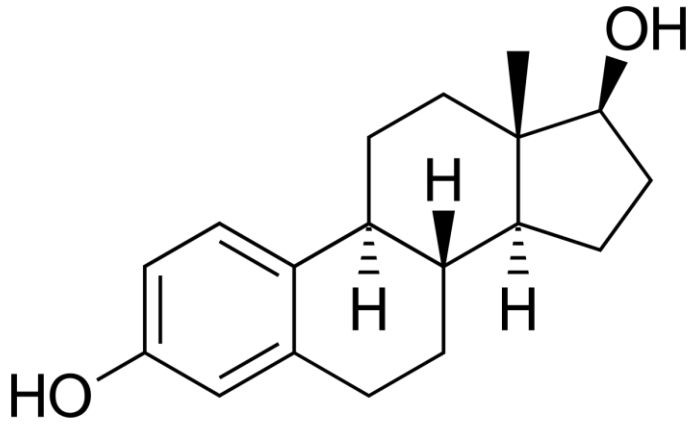
This is the basis of every **cell membrane**.

Steroids can pass through membranes

B1.1.13



Testosterone. NEUROtiker, public domain.



Oestradiol. NEUROtiker, public domain.

Steroids have **four fused rings**. They are mostly **non-polar**, so they dissolve into the membrane core and pass straight through.

So **oestradiol** and **testosterone** bind receptors **inside** their target cells.

Try it

Identify steroids in diagrams by the pattern of four rings.

Check-in: class 3

B1.1.8–13

1. State how many waters are released forming a triglyceride.
2. Distinguish saturated from unsaturated fatty acids.
3. Explain why triglycerides suit long-term energy storage.
4. Explain how phospholipids form a bilayer.
5. Explain why steroids can enter cells.

How to use this

Answer without your notes, then check against slides 19–27.

Common mistakes to avoid

Not this	But this
Condensation uses water.	It releases water; hydrolysis uses it.
Cellulose is made of alpha-glucose.	Beta-glucose; starch and glycogen use alpha.
A triglyceride is three glycerols and a fatty acid.	One glycerol and three fatty acids.
Unsaturated fats have more hydrogen.	Saturated ones hold the most hydrogen.
Phospholipid heads are hydrophobic.	Heads hydrophilic; tails hydrophobic.

Summary: carbohydrates and lipids

	Key idea
B1.1.1–3	Carbon: four bonds; condensation and hydrolysis
B1.1.4	Hexoses and pentoses; glucose properties
B1.1.5–6	Starch, glycogen (alpha); cellulose (beta)
B1.1.7	Glycoproteins: recognition, ABO
B1.1.8–11	Triglycerides: ester bonds, saturation, storage
B1.1.12	Phospholipid bilayers
B1.1.13	Steroids cross membranes

Check that you can...

1. **Distinguish** between condensation and hydrolysis
2. **Compare** starch, glycogen and cellulose
3. **Outline** the role of glycoproteins
4. **Draw** simplified triglycerides and phospholipids
5. **Explain** the effect of saturation on melting point
6. **Explain** bilayer formation and why steroids cross membranes

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the type of reaction that links two monosaccharides, and the molecule that it releases.
2. Give two ways in which the structure of glycogen suits it to energy storage.
3. Explain why a phospholipid is described as amphipathic.

Image credits

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Alpha-D-Glucose.svg – Yikrazuul – Public domain
Glycogen structure.svg – Mikael Häggström – Public domain
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